Furthermore, the composite nature of $F(h, y)$ provides a natural defense surrogate over-optimization (Goodhart's Law). Because the functional aggregates structurally distinct failure modes – logical contradiction, intent drift, miscalibration, and hazard – adversarial exploitation of any single component is likely to degrade performance on others. For instance, a policy that games the NLI classifier to avoid contradiction penalties by producing vacuous or generic text will incur penalties from the InfoGain or Value Conflict components. This structural diversity makes the surrogate objective more resistant to single-metric gaming than monolithic reward signals.

We do not claim pointwise bounds on the approximation error $|F^-(h, y) - \mathcal{R}_{\text{risk}}|$; rather, we require only that misorderings be sufficiently rare that the induced policy gradient remains aligned with the true risk gradient in expectation.

This places the framework squarely within the tradition of surrogate optimization in reinforcement learning, where unobservable objectives are optimized through structured, learnable proxies that preserve the ordering of actions by long-term utility. Under this view, the success of Frictive Policy Optimization does not depend on recovering the true epistemic risk functional, but on constructing a friction surrogate whose gradients induce interventions that systematically improve belief quality, reduce downstream failure, and stabilize long-horizon epistemic behavior.

In this sense, the friction functional plays the same conceptual role as learned reward models in preference-based reinforcement learning: it mediates between unobservable normative criteria and tractable learning signals, and its adequacy is ultimately validated not by calibration to ground truth, but by the quality of the policies it induces.

5.3 Interpretation.

The net friction functional $F(h, y) = F^+(h, y) - F^-(h, y)$ thus provides a structured, interpretable, and differentiable measure of epistemic quality. Each component corresponds to a distinct failure mode or benefit that can be measured, optimized, and audited independently.

To summarize the decomposition:

- **Productive friction** $F^+(h, y)$ should be *encouraged*: it captures deliberate epistemic interventions (clarification, verification, information seeking) that trade short-term interaction cost for long-term risk reduction.
- **Unproductive friction** $F^-(h, y)$ should be *suppressed*: it captures epistemic and normative failures (miscalibration, contradiction, hazard, silent intent fixing) that increase downstream risk without benefit.

This decomposition allows learning algorithms to distinguish between friction that should be encouraged and friction that should be suppressed—a distinction that cannot be captured by a single scalar measure of "helpfulness" or "compliance." In the FPO methods that follow, the net friction F enters reward shaping, preference pairing, and trajectory ranking with a positive sign, reflecting the principle that higher net friction corresponds to better epistemic quality.

6 Frictive Policy Optimization

In this section, we introduce a unified family of *Frictive Policy Optimization* (FPO) methods for learning epistemically aware intervention policies in dialogue. The central learning problem is not merely to generate fluent or helpful responses, but to decide *when to answer, when to clarify, when to refuse, and when to defer*, as a function of evolving epistemic risk.

Standard alignment methods implicitly assume that all actions are answers, and that epistemic regulation can be imposed externally through decoding heuristics, refusal filters, or post-hoc safety layers. In contrast, FPO treats intervention choice itself as a first-class control problem: the policy must learn not only *what* to say, but *whether* to speak, *how* to intervene, and *how strongly* to trade off short-term utility against long-horizon epistemic quality.

All methods in the FPO family instantiate the same theoretical commitment: that epistemic friction should be learned as part of the policy, rather than imposed as a heuristic or post-hoc filter. The methods differ in where friction enters the learning loop—through shaped rewards, preference supervision, trajectory-level ranking, or policy regularization—but they share a common objective: to induce policies whose intervention behavior is calibrated to epistemic risk.

The first three methods (FAR, FPP, and GRFR) differ in how epistemic friction enters the *learning signal*: through reward shaping, preference supervision, and group-relative ranking, respectively. Friction-Conditioned Trust Regions (FTR), introduced in Section 6.5, operate instead at the level of *policy regularization*, constraining how far the policy may deviate from a base model as a function of epistemic risk. As such, FTR is orthogonal to the other methods and can in principle be combined with any of them.

6.1 General Learning Framework

We now describe the general learning setting shared by all FPO methods. The goal is to learn a policy that maps dialogue histories not directly to surface responses, but to *intervention-structured actions* whose long-horizon epistemic consequences differ qualitatively. Unlike standard language model training, where each training example is a single input–output pair, FPO must learn a policy over a mixed action space consisting of both intervention types and their realizations.

We assume access to the following:

- A base policy π_0 (e.g., instruction-tuned or RLHF-trained);
- A dataset $\mathcal{D} = \{(h, y)\}$ of dialogue contexts and candidate responses;
- Optional preference annotations or groupings;
- Auxiliary estimators for components of $F(h, y)$.

As introduced in Section 4, we factor the policy as:

$$\pi_\theta(y \mid h) = \sum_{a \in \mathcal{A}} \pi_\theta(a \mid h)\, \pi_\theta(y \mid h, a) \tag{16}$$

where $\mathcal{A}$ is the taxonomy of frictive interventions. We define a finite intervention action space,

$$\mathcal{A} = \{\text{answer}, \text{clarify}, \text{verify}, \text{redirect}, \text{refuse}\},$$

corresponding to the core classes of frictive interventions introduced in Section 3. Here, *answer* is treated as a zero-friction action corresponding to ordinary response generation; including it explicitly in $\mathcal{A}$ allows the intervention policy to represent the choice *not* to intervene as a first-class decision, with $C_{\text{fric}}(\text{answer}) = 0$.

For notational and modeling simplicity, we treat *challenge* as an operational subtype of *verification* and *redirection*, since all three involve questioning or revising problematic premises. Accordingly, we do not introduce *challenge* as a separate atomic action in the formal action space $\mathcal{A}$, although it remains part of the underlying conceptual taxonomy in Section 3.

In the interest of space, we do not introduce meta-dialogue as a separate atomic action in the formal action space $\mathcal{A}$. Instead, such behaviors are subsumed under clarification and redirection, since they function operationally as higher-order forms of epistemic repair and intent negotiation.

This factorization makes explicit a distinction that is implicit in most alignment pipelines: the decision of *what kind of epistemic intervention to perform* is separated from the decision of *how to realize that intervention linguistically*. In standard training, these two decisions are conflated into a single generation step. In FPO, they are disentangled and jointly optimized. This induces two coupled learning problems:

1. learning an *intervention policy* $\pi_\theta(a \mid h)$,
2. learning a *realization policy* $\pi_\theta(y \mid h, a)$.

All FPO methods differ from standard alignment in that they explicitly train the intervention policy, rather than assuming that every action is an answer. This makes intervention choice itself a learned object, rather than a byproduct of decoding heuristics, refusal filters, or post-hoc safety layers.

From a learning-theoretic perspective, there are only a small number of principled ways to incorporate a structured surrogate such as $F(h, y)$ into policy learning. Friction may enter as an additive term in the reward, as a latent signal in preference supervision, as a ranking criterion over trajectories, or as a constraint on policy updates. The four methods introduced below—FAR, FPP, GRFR, and FTR—correspond exactly to these four design points.

Together, they define a minimal and complementary family of algorithms for learning risk-sensitive intervention policies, spanning reward-based, preference-based, ranking-based, and regularization-based alignment.

6.2 Friction-Augmented Rewards (FAR)

Friction-Augmented Rewards (FAR), originally introduced in Pustejovsky and Krishnaswamy (2025), extends standard reinforcement learning by adding a gated friction signal to the task reward. In the present formulation, FAR is generalized to incorporate an explicit risk-gating mechanism and a structured friction functional.

Objective. Friction-Augmented Rewards extends standard reinforcement learning by adding a gated *epistemic friction* signal to the task reward. This reward is designed to selectively encourage epistemic interventions only when they are justified by elevated risk, while explicitly penalizing their interaction cost.

$$R'(h,a,y) = R_{\text{task}}(h,a,y) + \alpha\, g(\mathsf{Risk}(h,y))\, F(h,y) - \beta\, C_{\text{fric}}(a) \tag{17}$$

where we assume:

- $g(\cdot)$ is a monotone gating function, e.g., $g(r) = \sigma(\kappa(r-\tau))$;
- $\mathsf{Risk}(h,y)$ is a pre-intervention estimate of epistemic risk used to gate friction, aggregating high-level risk indicators for the candidate intervention before it is executed in context h;
- $C_{\text{fric}}(a)$ penalizes intervention cost.

We emphasize that the two penalty terms in (17) play conceptually distinct roles. The term $C_{\text{fric}}(a)$ captures *pure interaction cost* (latency, user burden, and refusal penalties associated with taking an intervention action) independent of its epistemic outcome. In contrast, the friction functional $F(h,y)$ captures the *epistemic and normative quality of the resulting response*, including miscalibration, contradiction, hazard, and value conflict. Separating these terms prevents productive information-seeking from being conflated with surface interaction cost, and allows the policy to trade off *how costly it is to intervene* against *how beneficial the intervention is epistemically*.

This objective defines the shaped reward optimized by FAR. The agent maximizes the standard task utility R_{task}, but in addition receives a friction-dependent bonus $\alpha\, g(\mathsf{Risk}(h,y))\, F(h,y)$ that is activated only in high-risk contexts. The explicit penalty $-\beta C_{\text{fric}}(a)$ prevents the policy from overusing epistemic interventions in low-risk situations. Together, these terms implement the principle "use friction when it matters, but pay for it when you do."

The gating function $g(\mathsf{Risk}(h,y))$ implements *risk-conditional shaping*: friction is rewarded only when the estimated epistemic risk is high. When risk is low, $g \approx 0$ and friction is effectively ignored; when risk is high, $g \approx 1$ and productive friction is actively encouraged. This prevents degenerate policies that intervene habitually and enforces selective, calibrated intervention.

Policy Gradient. Any policy-gradient or actor–critic method may be used to optimize the shaped objective in (17) (Williams, 1992; Sutton et al., 2000; Schulman et al., 2017):

$$\nabla_\theta J_{\text{FAR}} = \mathbb{E}\left[\nabla_\theta \log \pi_\theta(a,y \mid h)\, R'(h,a,y)\right] \tag{18}$$

This is a standard policy-gradient update: it increases probability mass on intervention–utterance pairs (a,y) that yield higher shaped return R'. Epistemic behavior is shaped indirectly through the reward, rather than by explicitly optimizing a separate risk-reduction advantage or belief-state objective. In this sense, FAR treats epistemic regulation as a form of reward shaping: risk-sensitive intervention behavior is induced by modifying the scalar learning signal, without altering the structure of the policy class or the optimization algorithm.

Discussion. Friction-Augmented Rewards is the simplest member of the FPO family and is directly compatible with existing RLHF and actor–critic pipelines (Christiano et al., 2017; Ouyang et al., 2022). It requires no change to the policy architecture and treats epistemic regulation as a form of reward shaping.

The main advantage of FAR is its generality: any reinforcement learning algorithm that optimizes a scalar reward can incorporate friction with minimal modification. This makes FAR attractive as a baseline method and as a drop-in extension of standard alignment pipelines.

However, FAR inherits the known limitations of reward shaping. First, it is sensitive to reward scaling: inappropriate choices of α and β can lead either to excessive intervention or to complete suppression of epistemic behavior. Second, its effectiveness depends critically on the quality and calibration of the surrogate friction signal $F(h,y)$. Noise or bias in F is directly propagated into the learning signal, potentially destabilizing training.

More fundamentally, FAR shapes epistemic behavior only indirectly, through scalar reward. It does not explicitly reason about preferences over interventions, relative comparisons between trajectories, or long-horizon epistemic ordering. For this reason, FAR is best suited to settings in which a reasonably accurate risk estimator is available and intervention decisions are primarily local in time.

The remaining methods in the FPO family address these limitations by incorporating friction through preference supervision (FPP), group-relative ranking (GRFR), and policy regularization (FTR), which allow epistemic structure to enter learning in ways that are not reducible to reward shaping alone.